Spring Semester 2025–2026

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 4

To be covered during the examples class at 16:00 on Monday 27th April. These exercises do not count toward your final module mark, however it is a good idea to attempt these questions prior to the examples class.

1. Suppose that the process $\{X_t\}$ is defined

$$X_t = a + bt + s_t + Y_t$$

where a and b are constants, s_t is a seasonal factor such that $s_t = s_{t-12}$ and $\{Y_t\}$ is a 'noise' process (note, $\{Y_t\}$ is not necessarily a white noise process).

We define $\{W_t\}$ to be the process:

$$W_t = (1 - B)(1 - B^{12})X_t.$$

Determine the ACF of $\{W_t\}$ for the following cases:

(a) Where $\{Y_t\}$ is a white noise process with variance σ_z^2.

(b) Where $Y_t = (1 + \theta B)Z_t$, here $\{Z_t\}$ is a white noise process with variance σ_z^2.

2. Consider a (weakly) stationary $AR(p)$ process, $\{X_t\}$, written

$$X_t = \phi_1 X_{t-1} + \ldots + \phi_p X_{t-p} + Z_t.$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2.

(a) Determine the optimal k-step ahead predictor for $k = 1, 2, 3$, for each of the cases $p = 1, 2, 3$.

(b) For the $AR(1)$ process, what is the mean squared prediction error?

3. The process $\{X_t\}$ is an $ARIMA(1, 1, 0)$ process, defined

$$(1 - \phi B)\nabla X_t = Z_t$$

where $|\phi| < 1$ and $\{Z_t\}$ is a white noise process with variance σ_z^2.

Derive the optimal k-step ahead predictor of X_{n+k}, written $\hat{X}_{n+k}(n)$, for this process.

4. Recall the identity for the optimal k-step ahead predictor

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j} = \sum_{j=0}^{\infty} b_{j+k} Z_{n-j},$$

and define the functions

$$B(\xi) = \sum_{j=0}^{\infty} b_j \xi^j, \quad C_k(\xi) = \sum_{j=0}^{\infty} b_{j+k} \xi^j \text{ and } D_k(\xi) = \sum_{j=0}^{\infty} d_j(k) \xi^j.$$

(a) Justify the claim in the lectures that

$$C_k(\xi) = B(\xi) D_k(\xi).$$

(b) Why is this identity of importance in prediction?

5. Consider the $AR(3)$ process

$$X_t = 0.3X_{t-1} + 0.2X_{t-2} + 0.1X_{t-3} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2. Suppose that $X_1, \ldots, X_n$ have been observed. Determine the predicted values of X_t at $t = n+1$ and $t = n+2$.

6. Suppose that $\{X_t\}$ is an $MA(1)$ process, written

$$X_t = Z_t + \theta Z_{t-1}.$$

The linear predictor of X_{t+1} in terms of $X_t, X_{t-1}, \ldots$ is defined

$$\hat{X}_{t+1} = \sum_{r=0}^{\infty} a_r X_{t-r}.$$

Determine the coefficients $\{a_r\}$ such that $\mathbb{E}[(X_{t+1} - \hat{X}_{t+1})^2]$ is minimised. Hence determine the form of $\hat{X}_{t+1}$.